TangibleChannel: An Innovative Data Physicalization System for Visual Channel Education

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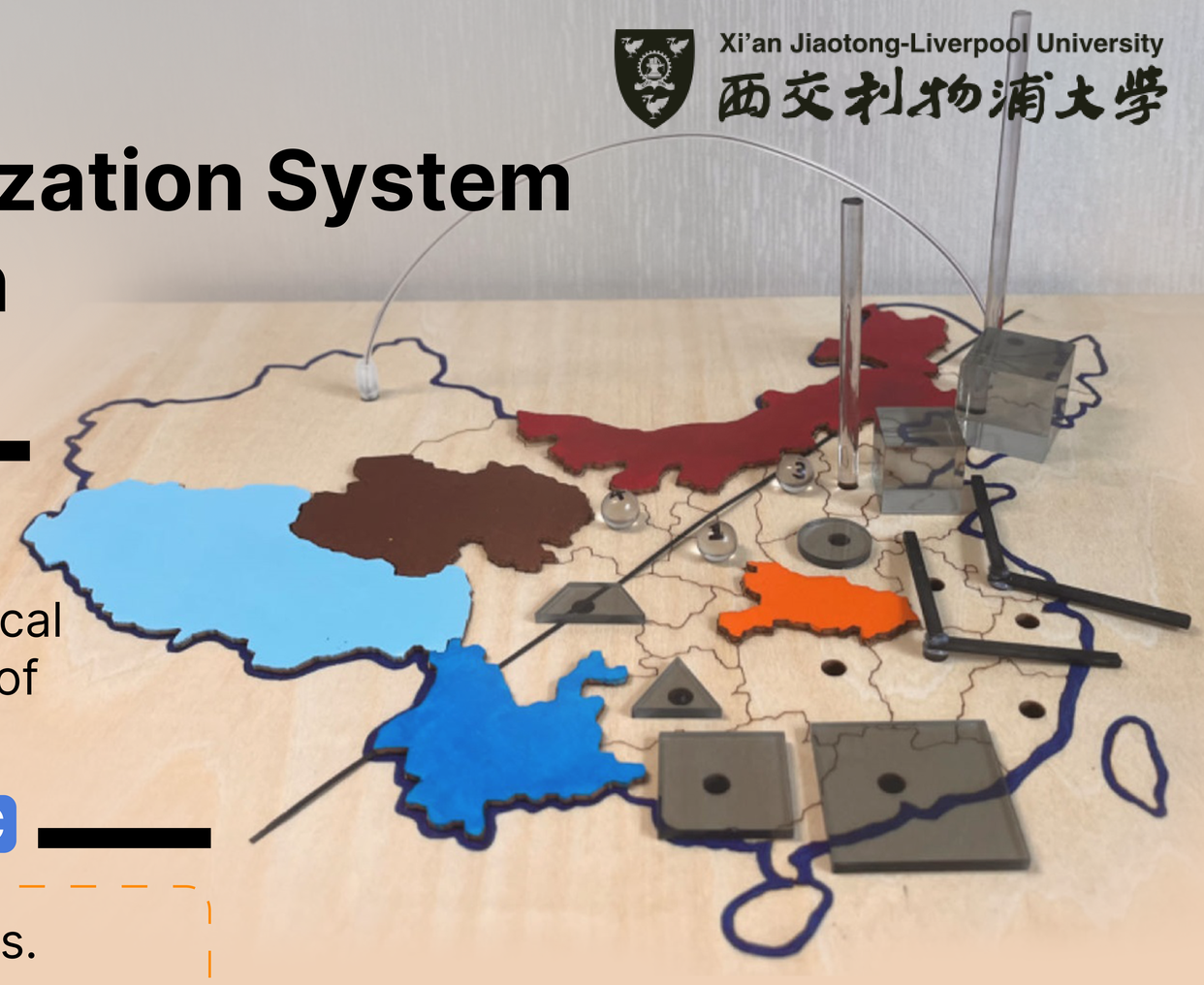


INTRODUCTION

Data physicalization is achieved by using physical objects and materials to convert digital data into tangible forms. We provide an overview of our physical pedagogical tools for enhancing the understanding of visual channels.

RESEARCH GOAL G & CONSIDERATION C

G1 Seclect suitable materials for visual channels.

G2 Uncover advantages of integrating physicalization into educational contexts.

G3 Explore value of data physicalization stems from its tangible form or the direct interaction.

C1 **Pedagogical Objectives:** Select appropriate visual channels for distinct data types.

C2 **Target Audience:** Tailor for students engaging visual channels and educators in the visualization domain.

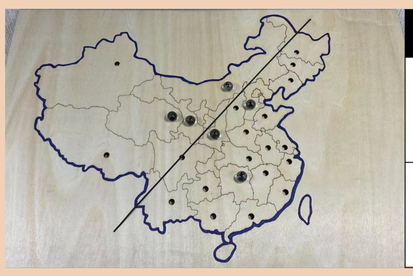
C3 **Data Quality:** Ensure incorporated data is both accurate and reliable.

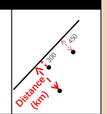
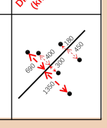
C4 **Interactivity:** Allow for exploration, comparison, and hands-on manipulation.

C5 **Aesthetic Appeal:** Create visually pleasing and harmonious representation.

SYSTEM PROTOTYPE

- TangibleChannel is crafted by laser-engraving the basic outlines of Chinese provinces onto a wooden board.
- To instantiate the coded **channels**, we employed a diverse range of materials M representing the data D of a distinct province.

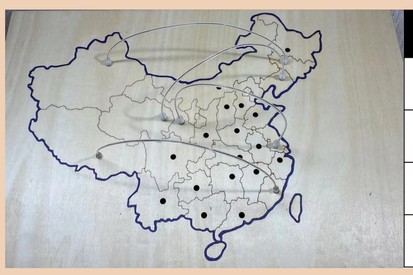


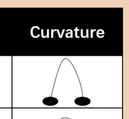
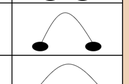
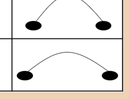
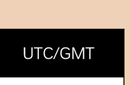
Distribution	Position
Aligned	
Unaligned	

Position (Aligned/ Unaligned)

M Epoxy ball

D The distance of balls from the Heihe-Tengchong line

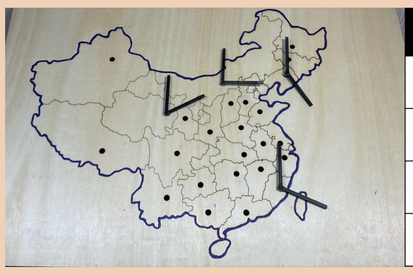


Distance (km)	Curvature
1729	
2489	
4255	
4279	

Curvature

M Plastic tube

D The distance between different cities by inserting pipes into both ends' holes



Angle	UTC/GMT
	+06:00
	+06:30
	+07:00
	+07:30

Angle

M Epoxy clock hand

D The timezone difference between the province and UTC

Color Hue	Region
	North China
	Northeastern
	East China
	South Central
	Southwest
	Northwest

Color Hue (ColorBrewer[1])

M Wooden shape

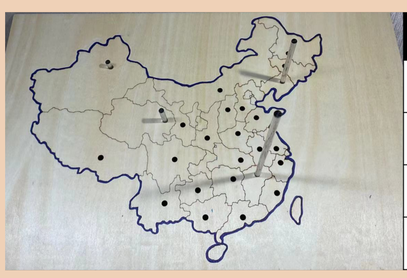
D The general region classification of Chinese provinces

Shape	Region
⬢	North China
⬢	Northeastern
●	East China
▲	South Central
▬	Southwest
▟	Northwest

Shape

M Epoxy token

D The general region classification of Chinese provinces

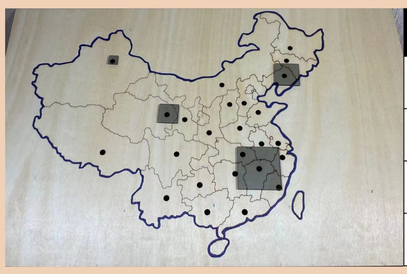


Side Length (cm)	Rainfall (mm)
1	100
3.4	337
8	800
15.6	1562

Length

M Epoxy stick

D The annual rainfall data for different provinces in China

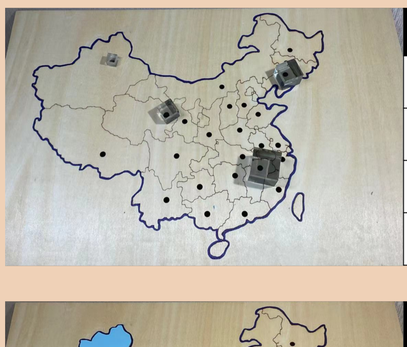


Side Length (cm)	Rainfall (mm)
1	100
1.8	337
2.8	800
3.9	1562

Area

M Epoxy token

D The annual rainfall data for different provinces in China



Side Length (cm)	Rainfall (mm)
1	100
1.5	337
2	800
2.5	1562

Volume

M Epoxy cube

D The annual rainfall data for different provinces in China

Color	Rainfall (mm)
	100
	337
	800
	1562

Luminance (Blue hue)

M Wooden shape

D The annual rainfall data for different provinces in China

CONCLUSION & FUTURE WORK

We introduced *TangibleChannel*, an educational system consisting of nine physical channels designed and fabricated to bolster student engagement and comprehension of visual coding concepts. In the future, we plan to explore the learning outcomes facilitated by our physicalization prototype compared to a screen-based education system.

[1] M. Harrower and C. A. Brewer. Colorbrewer. org: an online toolfor selecting colour schemes for maps. The Cartographic Journal,40(1):27–37, 200